are missing out on the opportunity to organize data more efficiently.

8.4.3 Modeling 2D Partitions with a Data Structure

Using the same, single direction approach for all points doesn’t really work… So, maybe dividing a plane into 4 quadrants would be a better idea?

Indeed, it works better than our previous attempts. Of course, since there are 4 quadrants, left and right partitioning doesn't apply any more.

We can use a tree where each node has 4 children instead of 2, one child for each possible quadrant. Points on the axes or on the lines passing by a point can be arbitrarily assigned to one of the quadrants (as long as this is done consistently).

This seems to work for $\mathbb{R}^2$, allowing us to overcome the main limit we had identified for our previous attempt: both x and y coordinates are taken into consideration to group points together (this would be even more apparent if you add more points to the diagram).